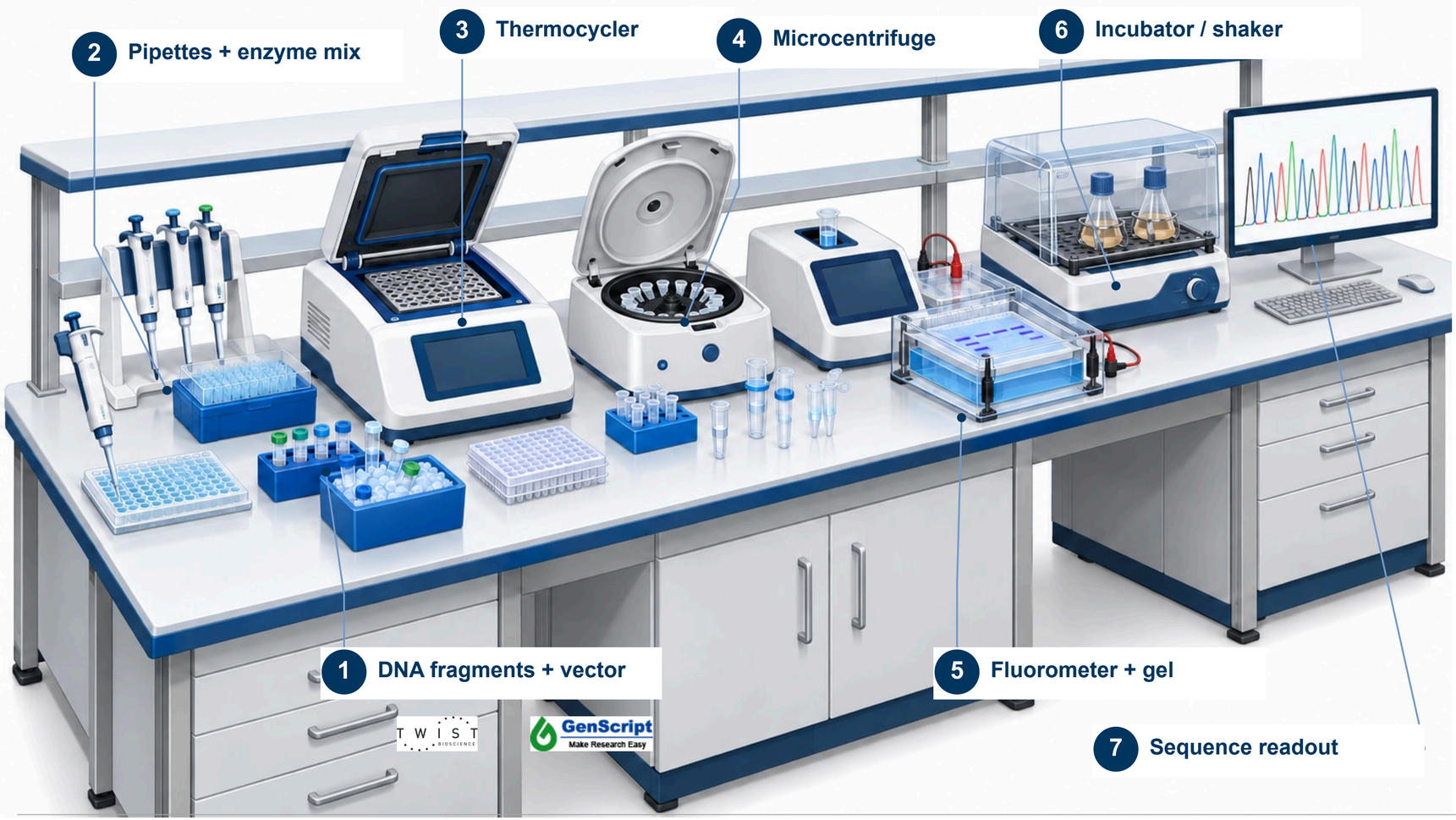
Inside a gene-building wet lab

An example bench for assembling DNA fragments into a sequence-verified plasmid

A working bench, shown through the tools on it



Tools, inputs and brands

1 DNA inputs

Oligos, fragments and vector DNA

Twist, GenScript, Tsingke

2 Reaction setup

Pipettes, tips, enzyme mix and tubes

Pipettes: Gilson / Eppendorf
Assembly mix: NEB / Vazyme

3 Thermal cycling

Thermocycler and PCR plates / tubes

Applied Biosystems, Bio-Rad

4 DNA cleanup

Microcentrifuge, columns and buffers

Centrifuge: Eppendorf
Kits: QIAGEN / TIANGEN

5 DNA quality checks

Fluorometer, gel tank and imaging

Qubit / Thermo Fisher, Bio-Rad

6 Clone growth

Shaking incubator, cells and cultureware

Shaker: Eppendorf
Cells / media: NEB / Thermo Fisher

7 Sequence confirmation

Sanger readout, service or sequencer

Service: Tsingke / GenScript
Instrument: Applied Biosystems

Recurring labware

Tips, PCR plates, microtubes, columns

Examples: Eppendorf / Thermo Fisher / JET Biofil

BIOFIL

Output: a verified DNA construct

Ready for downstream expression and assays

Generic illustration. Brands are alternatives. Numbers identify tool groups, not protocol steps. Sequencing may be outsourced. Sources: workflow/SOURCES.md.